Progress on the remaining weak-signal problems

Status: 2026-09-11. The work supplies a global computational benchmark and narrows the loading/adaptation questions. It does **not** solve all three general open problems. Earlier manuscripts and numerical results are preserved.

What is established

1. The ideal finite-net residual criterion is implemented on the complete continuous binary rank-one tensor ball. A global lower/upper objective bracket distinguishes an approximately optimal fit from a local solution. Confidence-set projection returns safe outer endpoints and feasible inner witnesses, with an explicit gap and budget status. This is an exponential benchmark, not a high-dimensional efficient optimizer.
2. For every loading tensor, fixed positive caps, and $M \le \tau$, minimax interval length is of order $M\rho(L)$ and squared risk is of order $M^2\rho(L)^2$, where $\rho(L)$ is the best unit product overlap. Computing that overlap need not be efficient. Unequal orthogonal weights additionally have an implemented optimized-prior mixture lower bound.
3. The existing ideal outer interval already adapts in minimax **rate** to every smaller positive fixed-cap class, without separation. This follows because all those classes share the rank-one lower bound. It does not resolve rank-dependent constants, growing caps, or pointwise oracle lengths.
4. A new supplied-cap tangent interval stays honest at zero and arbitrarily weak signals. Unresolved singular values receive a worst-case angle bound. Correctly calibrated artificial splitting and a simultaneous direct interval implement it in the original single-observation experiment. Strong signals can shorten it; weak-signal minimax optimality is unproved.

Complete statements and proofs are in theory/results.md and the technical supplement. The audit separates proved regimes, numerical certificates, and remaining requirements.

Newly run optimizer experiments

All supports and 1,002 calibrated scan constraints were used for order three and mode size two. The optimization parameter space is continuous. The stopping tolerance is 0.2τ, with 600 branch expansions per trial. Intervals reported below always use the outer bounds, including budget stops.

Signal	Loading	Covered / trials	Coverage 95% Wilson interval	Endpoint gap certified to tolerance	Mean outer length	Mean nodes
zero	flat	40/40	[0.912, 1.000]	37/40	3.148	348.5
zero	arbitrary	40/40	[0.912, 1.000]	39/40	3.999	188.1
weak_dense	flat	40/40	[0.912, 1.000]	40/40	3.154	326.3
weak_dense	arbitrary	40/40	[0.912, 1.000]	40/40	4.000	177.4
boundary_atom	flat	40/40	[0.912, 1.000]	37/40	3.112	349.2
boundary_atom	arbitrary	40/40	[0.912, 1.000]	38/40	3.991	202.7
strong_dense	flat	40/40	[0.912, 1.000]	0/40	2.867	600.0
strong_dense	arbitrary	40/40	[0.912, 1.000]	0/40	4.547	600.0

There are 320 new optimizer trials, with 231 meeting the endpoint tolerance. The strong-signal cases hit the short node budget; their wider outer intervals remain the reported result. Full coverage in these finite trials is not

a proof of uniform coverage. The small dimensions and conservative union thresholds do not establish a useful high-dimensional minimax gain.

Independent analytic checks include a one-atom residual minimum with known exact value, the exact structural support of a flat binary loading, a certified empty confidence set, and a positive non-rank-one residual optimum. The latter has global bracket [8.831631, 8.926524]. An intentional one-node stop verifies that incomplete endpoint optimization is not silently presented as a completed optimizer.

Newly run original-model adaptation experiments

Each trial starts with one $4 \times 4 \times 4$ Gaussian tensor at $\tau = 1$. Artificial splitting uses noise scale $\sqrt{2}$ in each half. The reported interval intersects the cap-valid interval at error $\alpha/2$ with the original-data direct interval at error $\alpha/2$, for $\alpha = 0.05$. The extra component, when present, lies on an orthogonal product atom.

Main amplitude	Weak extra	Supplied mode cap	Covered / 1000	Coverage 95% Wilson interval	Mean length	Tangent chosen	Previous separated method covered / 1000
0	0	1	979	[0.968, 0.986]	3.438	0	1000
0.5	0	1	970	[0.957, 0.979]	3.404	0	0
2	0	1	976	[0.965, 0.984]	3.171	0	0
2000	0	1	964	[0.951, 0.974]	4.337	0	958
2000	0.5	2	962	[0.948, 0.972]	4.343	0	962
2000	0	2	972	[0.960, 0.981]	4.348	0	959
200000	0	1	954	[0.939, 0.965]	3.018	1000	968
200000	0	2	955	[0.940, 0.966]	4.349	0	965

There are 8,000 new trials here. The old method is deliberately used outside its separation assumption in the weak-signal rows; those failures show why removing the assumption from the statement alone is invalid. The new interval retains coverage through conservative fallback. At a very large amplitude and a tight cap it uses the tangent option. A loose cap can prevent that gain even when the actual signal has rank one: the procedure does not pretend to have solved actual-rank adaptation. The very large amplitude also shows how conservative its finite-sample bias bound is.

Loading calculations and verification

The saved loading calculation has 27 combinations of support size, weight profile and budget. It evaluates a valid full-data lower bound, not a simulation or a proof of a matching general rate.

Fresh finite checks passed: 1,000 cosine ranges, 1,800 parameter-box containment points, 5 optimizer contract cases, 120 independent mixture identities, 250 first-order bias certificates, and 180 cap-valid tangent certificates including weak, zero, lower-rank, and strong cases.

The bounds are proved in real arithmetic. Implementations use double precision and conservative guards, not formal directed-rounding arithmetic. Run manifests record seeds, software versions and source hashes. The previous stage's tracked files are checked against its baseline hashes.

What remains open

- Efficient minimax weak-signal intervals at general dimensions and caps.
- Sharp larger-budget geometry for every loading and unequal weight sequence.
- Optimal growing-cap factors and pointwise adaptation where it is possible.
- General-dimensional certified optimization and formal numerical enclosures.

The code, proofs, and new experiments are a concrete advance, with those general problems still open. Reproduction commands are in README.md.